

By converting $x(t) = 1 - t^2$, $t \in [-1,1]$ by using the Fourier series the $b_3 =$

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cdot \cos(k\omega_0 t)) + \sum_{k=1}^{\infty} (b_k \sin(k\omega_0 t))$$

- ☐ a. -0.25
- ☐ b. 1.25
- ☐ c. 0.25
- ☒ d. 0
- ☐ e. -1.25

If $x(t) = 1 - t^2$, $t \in [-1,1]$ by using the Fourier series, the dc value equals to

☐ a. 1.5

☒ b. 0.667

☐ c. 0

☐ d. 0.5

[Clear my choice](#)



C.N.E

COMPUTER NETWORK
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By converting $x(t) = 1 - t^2$, $t \in [-1,1]$ by using the Fourier series the fundamental frequency component will be when $k =$

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cdot \cos(k\omega_0 t)) + \sum_{k=1}^{\infty} (b_k \sin(k\omega_0 t))$$

☒ a. 1

☐ b. 0

☐ c. 3

☐ d. 2

By converting $x(t) = 1 - t^2$, $t \in [-1,1]$ by using the Fourier series the $a_n =$

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(k\omega_0 t)) + \sum_{k=1}^{\infty} (b_k \sin(k\omega_0 t))$$

☐ a. all are correct

☐ b.

$$\frac{4(-1)^k}{\pi^2 k}$$

☒ c.

$$\frac{-4(-1)^k}{\pi^2 k^2}$$

☐ d.

$$\frac{-4(-1)^k}{\pi k^2}$$

☐ e.

$$\frac{4(-1)^k}{\pi k}$$

By converting $x(t) = 1 - t^2$, $t \in [-1,1]$ by using the Fourier series the $a_1 =$

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cdot \cos(k\omega_0 t)) + \sum_{k=1}^{\infty} (b_k \sin(k\omega_0 t))$$

☐ a.

$$\frac{-4}{\pi^2}$$

☒ b.

$$\frac{4}{\pi^2}$$

☐ c.

$$\frac{-4}{\pi}$$

☐ d.

$$\frac{4}{\pi}$$

☐ e. none of the all

$$\text{If } X(\omega) = \frac{2e^{j6t}}{j\omega} \text{ so } x(t) =$$

☐ a.

$$-u(t-6) + u(-t+6)$$

☐ b.

$$u(t-6) - u(-t-6)$$

☐ c.

$$u(t-6) + u(t+6)$$

☒ d.

$$u(t-6) - u(-t+6)$$

If $X(\omega) = \frac{4}{-j\omega}$ so $x(t) =$

☐ a.

$$2(u(t))^2$$

☐ b.

$$2u(t) - 2u(-t)$$

☐ c.

$$-2u(t) + 2u(-t)$$

☒ d.

$$-2\text{Sng}(t)$$

If $f(t) = e^{-5|t|}$ @ $t = 0$, so that $F(1) =$

- ☐ a. none of the all
- ☐ b. j
- ☒ c. 1
- ☐ d. $-j$
- ☐ e. -1

[Clear my choice](#)

If $f(t) = e^{-b|t|}$ where $b > 0$ so the $F(\omega) =$

☐ a.

$$F(\omega) = \frac{2a}{2a^2 + \omega^2}$$

☒ b.

$$F(\omega) = \frac{2a}{a^2 + \omega^2}$$

☐ c.

$$F(\omega) = \frac{2a}{2a + \omega}$$

☐ d.

$$F(\omega) = \frac{a}{a + \omega}$$

Clear my choice

If $f(t) = e^{-b|t|}$ where $b > 0$ so the $F(\omega) =$

☐ a.

$$F(\omega) = \frac{2a}{2a^2 + \omega^2}$$

☒ b.

$$F(\omega) = \frac{2a}{a^2 + \omega^2}$$

☐ c.

$$F(\omega) = \frac{2a}{2a + \omega}$$

☐ d.

$$F(\omega) = \frac{a}{a + \omega}$$

Clear my choice



Time left 0:11:32

Question **3**

Not yet answered

Marked out of 1.00

Flag question

Non-periodic and deterministic signals are energy signals whereas periodic and random signals are power signals.

Select one:

- ☐ True
- ☐ False

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2022-2-ELE0321-SIGNALS AND SYSTEMS

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If $x(t)$ has fundamental period $T=10$, then $x(5t-8)$ has fundamental period

- ☐ a. 5
- ☒ b. 2
- ☐ c. 25
- ☐ d. 0.5
- ☐ e. 15

[Clear my choice](#)

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Question 1

Time left 0:13:48

Not yet answered

Marked out of 1.00

Flag question

Find the fundamental period and frequency of the following signal

$$x(t)=2\cos(6(\pi)t+\pi/6)$$

- ☐ a. $T=0.33$ sec , $f=3$ Hz
- ☐ b. $T=10$ sec , $f=0.1$ Hz
- ☐ c. $T=5$ sec , $f=0.2$ Hz
- ☐ d. $T=0.16$ sec , $f=6$ Hz

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カエル 🐸 🐸

now

Telecommunications Engineering

بدنا مين يساعدنا

Time left 0:14:29

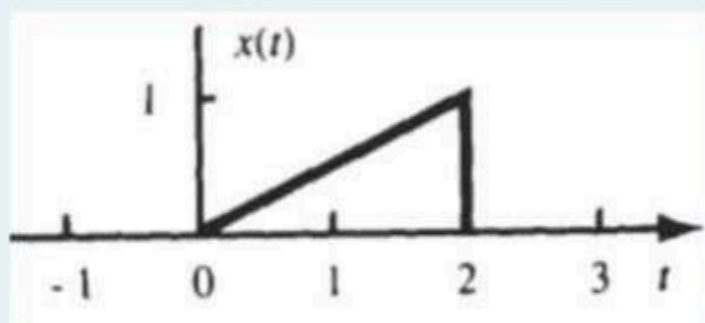
Question 1

Not yet answered

Marked out of 1.00

Flag question

the following signal can be expressed as



- ☐ a. $t [u(t)-u(t-2)]$
- ☐ b. $tu(t)-u(t-2)$
- ☐ c. $0.5t [u(t)-u(t-2)]$
- ☐ d. $2t [u(t)-u(t-2)]$

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Textbook and References ▶

الوقت المتبقي 0:13:27

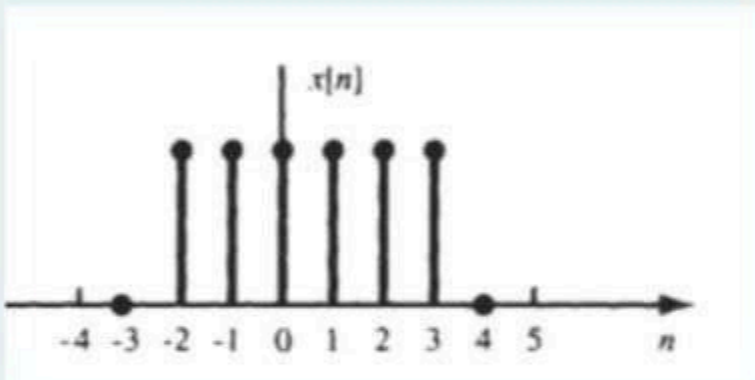
سؤال 1

غير مجاب عليه بعد

الدرجة من 1.00

٣ علم هذا السؤال

The $x[n]$ amplitude is one, so for the following signal drawing is



$x[n] = u[n+2] - u[n-4]$

a ☐

$x[n] = u[n-2] - u[n+3]$

b ☐

$x[n] = u[n-2] - u[n+4]$

c ☐

$x[n] = u[n] - u[n-4]$

d ☐

$x[n] = u[n+2] - u[n-3]$

e ☐

الصفحة التالية

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